

MS101 2023-24/II (Spring)

EE Quiz 2 - (Mar 22, 2024) - Numerical Answers

Question	Answer	Comments
1	Solution 1: 180 RPM = 3 RPS = 12 sectors/sec Waveform period: 83.33 ms Solution 1: 180 RPM = 3 RPS = 24 sectors/sec Waveform period: 41.67 ms	Full credit to both answers.
2	0, 1 (A, B)	
3	1 (B)	
4	$I_C = 3 \text{ mA}$, $I_B = 0.03 \text{ mA}$, $R = 180 \text{ k}\Omega$	
5	$I_C = 1.9753 \text{ mA}$	
6	0 (A)	
7	1 (B)	
8	1 (B)	
9	1, 3 (B, D)	
10	1, 3 (B, D)	
11	Current through relay = 24.5 mA	
12	SAFE: 1, 3 (B, D)	